

LIME

Local Interpretable
Model-agnostic Explanations

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ACT 1 Build

The method, taught properly: what LIME optimizes and the six steps that implement it.

ACT 2 Break

Three measurements on our own run that undo part of what Act 1 just taught.

ACT 3 Rebuild

What survives the breaking, and the rules that follow for using it.

→ One dataset throughout

569 breast-cancer patients, one RandomForest, one patient explained.

ACT 1

Build

LIME as its authors describe it, and as the package implements it. We will take it entirely at face value until Act 2.

Four words, four commitments.

The first is the one people skip: one prediction at a time, never the whole model. And the explanation may use features the model never saw — train on PCA components, explain in the original survey questions.

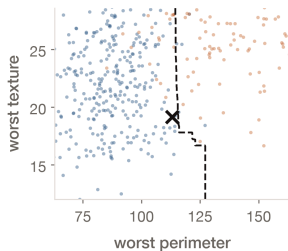
LOCAL**faithful near one point****INTERPRETABLE****readable by a human****MODEL-AGNOSTIC****any model, any data****EXPLANATIONS****input → prediction, legible**

One equation, three demands.

$$\xi(x) = \arg \min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

- 01 Faithful to the black box — $\mathcal{L}$ against f , a RandomForest of 300 trees
- 02 Faithful *here* — $\pi_x = \sqrt{\exp(-d^2/\nu^2)}$, width $\nu = 0.75\sqrt{30} \approx 4.11$
- 03 Readable — Ω : 8 of 30 features, and g is a Ridge with $\alpha = 1$

THE PICTURE



● benign ● malignant x patient #67
Dashed line: the forest's own decision boundary in this slice.

THE CODE

```
model = RandomForestClassifier(  
    n_estimators=300,  
    min_samples_leaf=3,  
)  
model.fit(X_train, y_train)  
  
# the only thing we ever  
# ask of f, all deck long  
p = model.predict_proba(X)[:, 1]
```

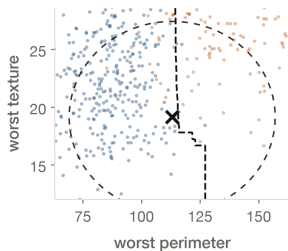
THE IDEA

Explain one prediction, not the model.

We may only call `f`. Patient #67 is used throughout: she sits on the boundary so the geometry stays visible, and the axes are her rank-1 and rank-3 features.

SO FAR — a model, and one row.

THE PICTURE



● benign ● malignant × patient #67
Dashed ring: the kernel's 0.95-weight contour, at 1.32σ .

THE CODE

```
# one width per feature,  
# each on its own scale  
mu = X_train.mean(axis=0)  
sigma = X_train.std(axis=0)  
  
# LIME's default kernel width  
# lime_tabular.py:243  
nu = 0.75 * np.sqrt(30)
```

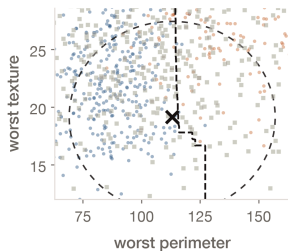
THE IDEA

Locality must be defined before it means anything.

A distance and a width, chosen by us, not by the data. The ring is the only kernel contour that fits this frame — and it is the 0.95 one. Act 2 shows where the half-weight contour actually falls.

SO FAR — + a definition of "near".

THE PICTURE



● benign ● malignant × patient #67
■ synthetic neighbor
All 30 features resampled, not just these two.

THE CODE

```
# sample_around_instance=False  
# is the default, so the cloud  
# is centred on the DATASET,  
# not on the patient  
Z = mu + sigma * rng.normal(  
    0, 1, size=(5000, 30))  
  
Zs = (Z - mu) / sigma
```

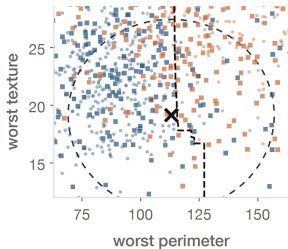
THE IDEA

**If you cannot look inside,
build data to probe with.**

This is where the assumptions enter.
Each of the 30 features is drawn
independently from its own marginal, and
the cloud is centred on the dataset rather
than on her.

SO FAR — + 5,000 probe rows.

THE PICTURE



● benign ● malignant × patient #67
■ pred. benign ■ pred. malignant
Square colour is the model's prediction, not a true label.

THE CODE

```
# the one moment the real  
# model is queried at all —  
# 5,000 rows of 30 numbers  
fz = model.predict_proba(Z)[: , 1]  
  
# note: raw Z, not Zs. The model  
# was trained on raw units.
```

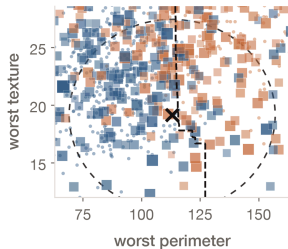
THE IDEA

The black box becomes its own teacher.

We now hold a supervised dataset LIME can learn from. Notice the squares do not separate along the drawn boundary — 68% against a 53% baseline — because each carries random values on the other 28 features.

SO FAR — + the model's answer on each.

THE PICTURE



● benign ● malignant × patient #67
■ pred. benign ■ pred. malignant
Square size and opacity encode the proximity weight π .

THE CODE

```
# distance over all 30  
# STANDARDIZED features,  
# not the two we plot  
d = np.linalg.norm(  
    Zs - x_scaled, axis=1)  
  
w = np.sqrt(  
    np.exp(-d**2 / nu**2))
```

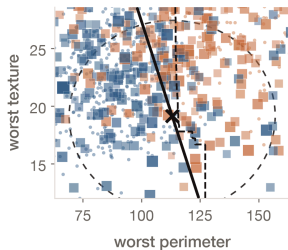
THE IDEA

This is the only step that makes the fit local.

Everything before it was global. The kernel is what is supposed to concentrate the regression near the patient — so watch how little the weights actually vary.

SO FAR — + a weight per row.

THE PICTURE



● benign ● malignant × patient #67
■ pred. benign ■ pred. malignant
Solid line: the local fit, drawn through the patient.

THE CODE

```
# stage 1 — pick 8 of 30,  
# itself weighted by w  
aux = Ridge(alpha=0.01).fit(  
    Zs, fz, sample_weight=w)  
top8 = np.argsort(  
    -abs(aux.coef_ * x_scaled))[:8]  
  
# stage 2 — the explanation  
g = Ridge(alpha=1).fit(  
    Zs[:, top8], fz,  
    sample_weight=w)
```

THE IDEA

The fitted model is the explanation.

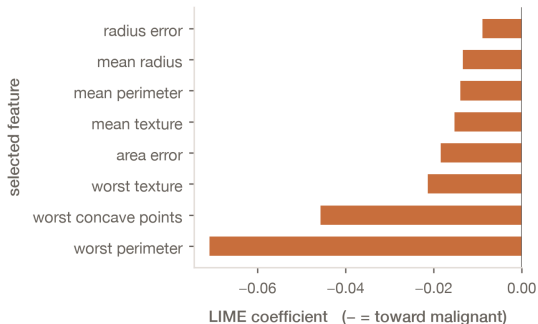
Two weighted stages: select 8 of 30, then fit. Its coefficients are what you report; the line drawn here is that same fit seen in the plane, through the patient rather than at $P=0.5$.

DONE — 8 coefficients, and $R^2 = 0.36$.

What the library actually hands you.

- 01 What a practitioner reads is a ranked list, never the geometry we just drew
- 02 Here it is unanimous: all eight selected measurements say malignant
- 03 Yet the model says benign — so the verdict lives in the 22 features LIME dropped

All eight selected features point the same way



ACT 2

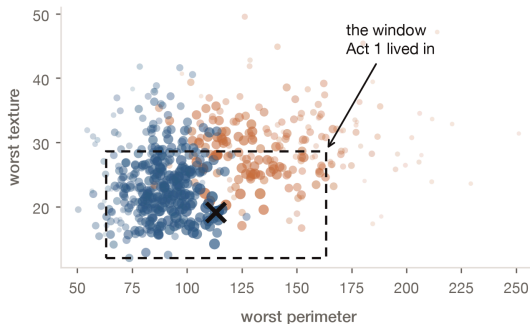
Break

Three claims about LIME, checked against our own run. Two are already in the literature and we verify them. The third is ours — and it does not survive contact with a second dataset.

The "local" fit is barely local.

- 01 Garreau & von Luxburg (2020) prove it: at a wide bandwidth the kernel is "equivalent to give weight 1 to every perturbed sample"
- 02 We verify it — deleting the kernel moves $g(x)$ by a median 0.002 and keeps all 8 features here, and the selection survives 92–100% across four datasets and three model classes
- 03 Its half-weight contour sits at 4.84σ — so 42% of all 569 patients carry weight above 0.5
- 04 And it is deliberate: they *recommend* the wide default, because a narrow one drives every coefficient to zero

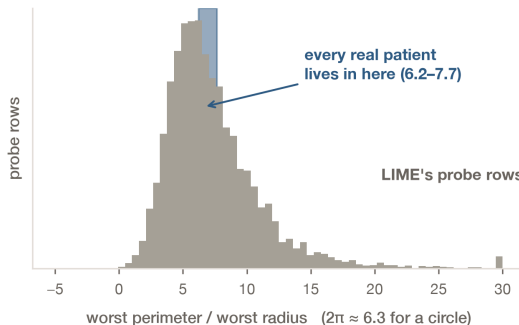
Dot size is each patient's actual kernel weight



It explains data that cannot exist.

- 01 76% of synthetic neighbours carry a negative measurement; perimeter/radius spans 1.8–21.5 (1st–99th pct) against a real 6.2–7.7
- 02 Because those perturbations are detectable as out-of-distribution, Slack et al. (2020) hid a racist classifier from LIME
- 03 Molnar lists this limitation — we are quantifying it, not discovering it

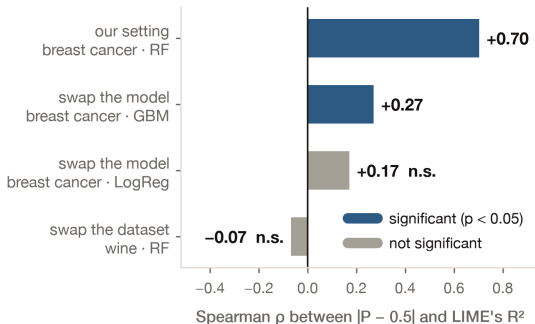
Three quarters of the probe rows are impossible patients



And one we measured that does not hold up.

- 01 On this dataset with this model, R^2 tracks the prediction's confidence: $\rho = +0.70$
- 02 Swap the model and it weakens. Swap in an unrelated dataset — *wine*, 13 chemical measurements, nothing to do with cancer — and it vanishes
- 03 A published evaluation finds *no* consistent pattern — and one setting where it runs the other way (Velmurugan et al., 2020)

It survives only the setting we happened to study first



ACT 3

Rebuild

Nothing above says throw LIME away. It says know which half of its answer to read — and that half is measurable.

A linear fit gives you two things. Only one of them holds.

DISCARD

The *level*: $g(x) = 0.497$ while $f(x) = 0.581$.

KEEP

The *direction*: 4° from where the model actually changes.

COSINE 0.997 IN-PLANE · 0.57 ACROSS 30-D

01 Read the direction, not the level

The sign and rank of the leading coefficients. Never the surrogate's own predicted probability.

02 Never rank explanations by R^2

It rewards saturated regions. Report it, but as a caveat, not a score.

03 Check $g(x)$ against $f(x)$

One line of code. If they disagree, the level is worthless — and often they do.

04 Re-run before you believe a ranking

The leading feature holds; the tail moves. Trust what survives the reseed.

Thanks!



github.com/wbendinelli/interpretable-ml-lectures

REFERENCES

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